Why Electronics Use Sine Waves

01 What is a sine wave?



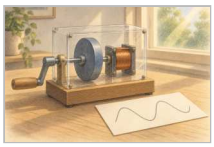
A gentle swing slows near each turn.

Look closer. Notice how the swing slows near the end of its trip.

A little explanation. A sine wave is a smooth back-and-forth wiggle that repeats. It can show how a position or an electric push changes over time. A gentle swing moves in a sine-like rhythm; its seat travels along an arc.

Try this. With Big nearby, take a gentle swing, bounce a soft toy spring over a tray, or move a rope from side to side on the floor. Draw the smooth wiggle you notice. Keep wall power and plugged-in devices closed and untouched; these are activities with a swing, spring, rope, paper, or an intact battery speaker.

02 How can spinning make a wave?



Spinning can make a smoothly changing electric push.

Look closer. Find the turning disk beside the wire coil.

A little explanation. A generator turns motion into electrical energy. As a magnet spins beside wire, the electric push changes smoothly, first one way and then the other. Many generators make the AC power supplied at home. AC means that the push changes direction repeatedly.

Try this. Draw a dot on a paper circle. Turn the circle slowly and watch the dot rise and fall.

03 What wiggles inside a speaker?



A speaker moves air to make sound.

Look closer. Look at the circle on the front of the speaker.

A little explanation. The speaker cone moves in and out, making the air nearby squeeze and spread. Those changes travel to your ears as sound. For a pure, steady tone, the cone moves in a sine-like rhythm. Music and voices make more complicated wiggles.

Try this. Let Big set an intact battery speaker to low volume. Rest fingertips lightly on its outer case, leaving the cone alone, and notice any quiet buzz. A drawing of the cone's motion works too.

04 Why use a smooth wiggle?



A sine wave is a smooth, repeating wiggle.

Look closer. Follow the rounded turns in Little's notebook.

A little explanation. Spinning parts can make sine waves naturally. Smooth changes help many motors and speakers work evenly. Sudden jumps can make extra buzz or jerky starts. Electronics also use steady power and other signal shapes; not every signal is a sine wave.

Try this. Draw a smooth wave and a zigzag. Trace them slowly with a finger and compare the turns.